Moo-tion: A Computer-Vision-Based System for Dairy Farming

Project Specification Document

Mert Özkaya	22002460
Muhammed Enes Şahin	22102919
Canberk Özba	22101815
Ahmet Burak Kılıçkaya	22103833
Emir Eroğlu	22202233

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1 Introduction

1.1 Description

Moo-tion is a computer-vision-based decision support system for automated estrus (heat) detection in dairy cattle using only existing farm security cameras. Instead of requiring invasive or expensive wearable sensors, Moo-tion passively monitors cows in their natural environment and analyzes their behavior through video streams.

The core idea is to continuously process the camera feed, detect individual cows, track their movement patterns, and identify behavior indicative of estrus (e.g., increased activity, mounting behavior, restlessness). When the system detects a cow that is likely to be in heat, it sends a notification to the farmer via a dedicated mobile application. The mobile application also allows real-time access to the live camera feed, historical alerts, and simple analytics (e.g., frequency of estrus events per animal).

By providing a purely camera-based and non-invasive solution, Moo-tion aims to reduce missed estrus events, optimize insemination timing, and ultimately improve herd fertility and farm profitability, while minimizing additional hardware costs and deployment complexity.

1.2 High Level System Architecture & Components of Proposed Solution

At a high level, Moo-tion follows a modular, service-oriented architecture composed of the following main components:

- **Camera Infrastructure**: One or more fixed security cameras installed in the barn or open paddock, providing continuous video streams. The system is designed to reuse existing IP cameras where possible (e.g., RTSP streams).
- **Video Ingestion & Preprocessing Module**: A streaming service that connects to the camera(s), decodes the incoming video, performs basic preprocessing (resolution scaling, frame sampling, background filtering), and forwards frames to the analysis pipeline. The system could also convert night images into day-like images for improved analysis.
- **Computer Vision & Estrus Detection Module**: A core analysis service that:
  1. Detects and localizes cows in each frame using object detection models,
  2. Tracks identified cows across frames to maintain individual trajectories,
  3. Extracts behavioral and motion features (e.g., speed, acceleration, mounting interactions),
  4. Applies a machine learning model to estimate the probability of estrus for each cow.
- **Backend API & Data Storage**: A server-side component that stores detections, alerts, and metadata in a database (e.g., time of detection, camera ID, cow ID if available) and exposes RESTful APIs to the mobile application.
- **Notification Service**: A lightweight service that monitors estrus probabilities and triggers push notifications or SMS messages to the farmer when certain thresholds are exceeded, optionally including snapshots from the video frame.
- **Mobile Application**: A cross-platform mobile app (e.g., Android and/or iOS) that allows users to:
  - View the live camera stream,
  - See active and historical estrus alerts,
  - Inspect corresponding video thumbnails or short clips,
  - Manage basic settings such as notification thresholds and camera sources.

A conceptual representation of the architecture is shown in Figure 1. The design enables deployment either fully on-premise (edge device near the barn) or in a hybrid mode where video analysis is done locally while the backend and notification services are hosted in the cloud.

Figure 1 illustrates the overall architecture of the Moo-tion system. The existing farm security camera continuously streams video to the Moo-tion server, where the video ingestion and preprocessing module prepares frames for the computer vision and estrus detection module. Detected events and alerts are

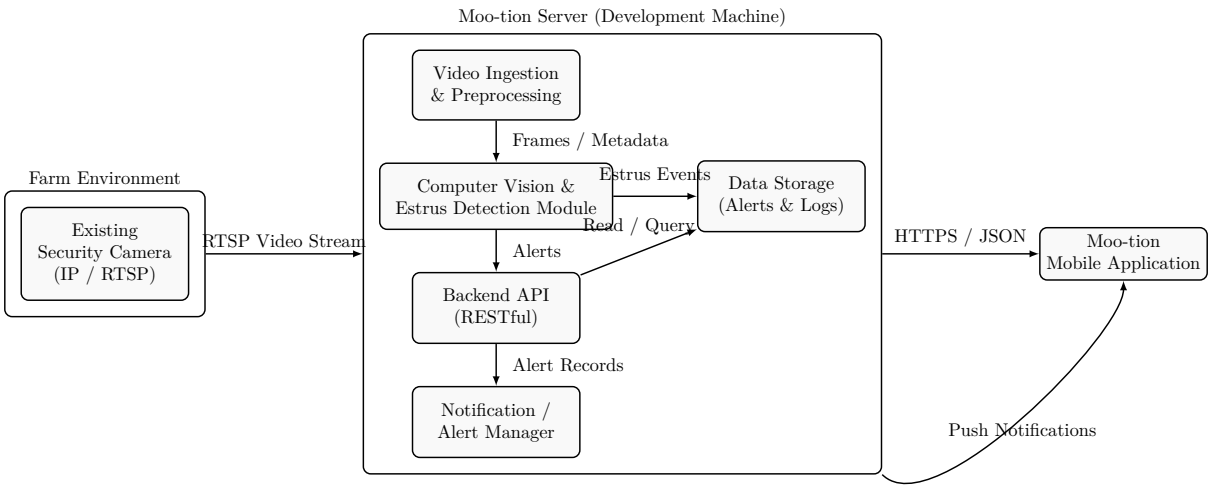

Figure 1: High-level system architecture of the Moo-tion estrus detection system.

stored in the data storage component and exposed through a RESTful backend API, while the notification manager triggers push notifications when estrus is detected. The mobile application communicates with the backend over HTTPS/JSON, allowing the farmer to monitor the system and receive alerts without needing any additional on-farm compute resources.

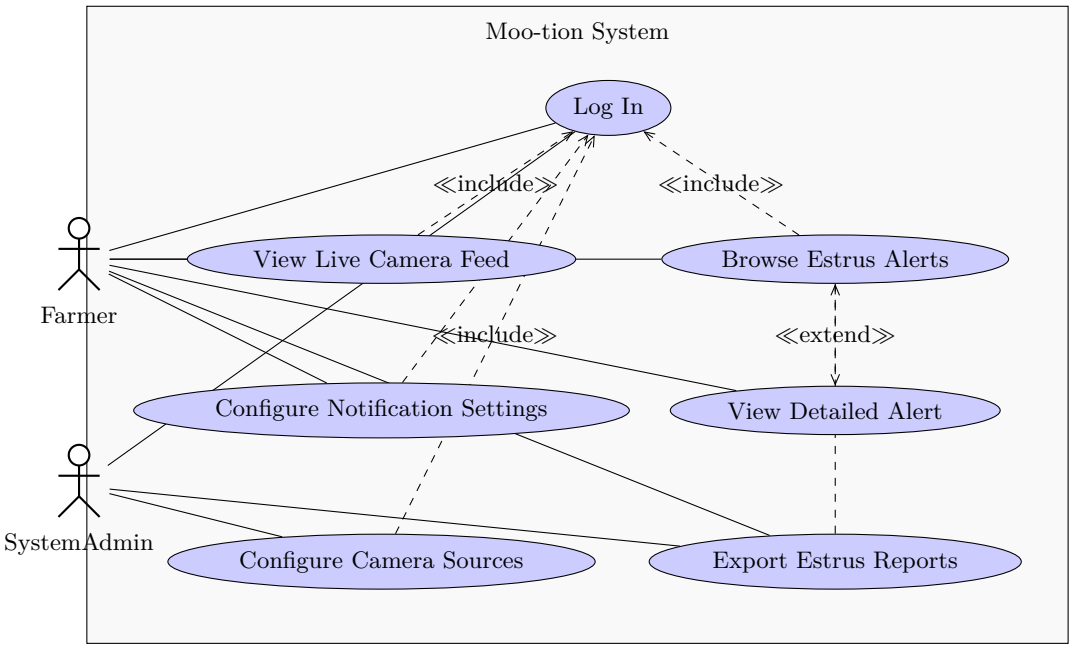

Figure 2: Use case diagram of the Moo-tion estrus detection system with include and extend relationships.

Figure 2 shows the main use cases of the Moo-tion system from the perspective of two actors: the Farmer and the System Administrator. Both actors must first execute the *Log In* use case, which is included in other use cases such as viewing the live camera feed, browsing estrus alerts, configuring notification settings, and managing camera sources. The *Browse Estrus Alerts* use case is further extended by *View Detailed Alert*, which represents an optional, more detailed inspection of a selected alert. Exporting estrus reports reuses the browsing functionality via an «include» relationship, reflecting that report generation is based on previously stored and filterable alert data.

1.3 Constraints

The design and implementation of Moo-tion are subject to several constraints that arise from the target deployment environment (farms), available hardware, economic considerations, and ethical responsibilities. These constraints directly influence the choice of algorithms, system architecture, and deployment strategy.

1.3.1 Implementation Constraints

From an implementation perspective, the key constraints are:

- **Network Connectivity**: Internet connectivity in rural areas may be intermittent or bandwidth-limited. The system must tolerate temporary disconnections and avoid continuous high-bandwidth uploads (e.g., avoid streaming raw video to the cloud in real time whenever possible).
- **Legacy Camera Infrastructure**: The solution is required to work with existing camera infrastructure (e.g., fixed IP cameras with RTSP), which constrains frame rate, resolution, and viewing angles. Algorithms must be robust to varying lighting conditions, occlusions, and camera viewpoints.
- **Real-Time Requirements**: Estrus detection does not need millisecond-level latency, but alerts should be generated within a practically useful time window (e.g., within minutes) to be actionable. This imposes soft real-time processing constraints on the video analysis pipeline.
- **Data Availability for Training**: High-quality labeled video data for estrus-related behaviors may be limited. The system design must allow for incremental model improvement and potential semi-supervised or transfer learning approaches.

1.3.2 Economic Constraints

Moo-tion is intended to be an affordable solution for small and medium-sized farms. This leads to several economic constraints:

- **Minimal Additional Hardware**: The system should reuse existing security cameras whenever possible and avoid dedicated wearable sensors or expensive hardware add-ons, reducing upfront capital expenditure.
- **Low Operational Costs**: Continuous cloud processing of high-resolution video can be costly. The design favors on-premise or edge processing to minimize recurring cloud compute and storage costs.
- **Scalable Licensing Model**: For commercial viability, the software should support a pricing model that scales reasonably with the number of cameras and herd size, making it accessible to farms with different budgets.
- **Maintenance Overhead**: The system should require minimal on-site technical support. Automatic updates, remote monitoring capabilities, and a simple user interface help reduce maintenance costs.

1.3.3 Ethical Constraints

The project must respect both animal welfare and human privacy, which introduces ethical constraints:

- **Animal Welfare**: Moo-tion must not encourage harmful or overly intrusive management practices. The system should be used as a decision-support tool, not as a replacement for veterinary expertise or responsible livestock management.
- **Human Privacy**: Security cameras may occasionally capture farm workers or visitors. Video storage and processing must comply with relevant data protection regulations, and access to video feeds should be restricted and audited.
- **Data Ownership**: Farms should retain ownership of their data (videos, labels, alerts). Any use of data for model improvement or research must be based on explicit consent.

1.4 Professional and Ethical Issues

Developing Moo-tion involves several professional and ethical responsibilities typically associated with software engineering and AI systems:

- **Software Quality and Reliability**: The system must be developed following good software engineering practices, including requirements analysis, version control, testing, and documentation, to ensure reliability in real farm conditions.
- **Security and Access Control**: Authentication, authorization, and secure communication (e.g., HTTPS, secure storage of credentials) are necessary to prevent unauthorized access to live camera feeds and sensitive farm data.
- **Transparency and Explainability**: While the underlying models may be complex, the system should provide interpretable feedback to farmers, such as highlighting the time interval and basic indicators that triggered an estrus alert.
- **Responsibility and Liability**: Clear disclaimers and usage guidelines must be provided to ensure that farmers understand the system as an assistive tool. Responsibility for final decisions regarding animal health and reproduction must remain with qualified personnel.
- **Sustainability**: Efficient resource usage (e.g., energy-aware processing on edge devices) should be considered, aligning with broader sustainability goals for agricultural technology.

1.5 Standards

The design and implementation of Moo-tion will take into account relevant technical and regulatory standards, including but not limited to:

- **Video and Streaming Standards**: Use of common protocols and formats such as RTSP for video streaming and H.264/H.265 for video encoding to ensure compatibility with off-the-shelf security cameras.
- **Software Quality Standards**: Adherence to general software quality principles inspired by standards such as ISO/IEC 25010, focusing on reliability, usability, maintainability, and security.
- **Information Security Standards**: Alignment with best practices from standards like ISO/IEC 27001 for secure handling of data, including access control, logging, and backup strategies.
- **Data Protection and Privacy Regulations**: Compliance with applicable data protection regulations (e.g., GDPR-like principles or local equivalents) concerning storage, processing, and sharing of video data that may contain identifiable individuals.
- **Mobile Platform Guidelines**: Conformance with Android/iOS platform guidelines for permissions (e.g., notification permissions, network access) and secure handling of user credentials and API keys.

2 Design Requirements

2.1 Functional Requirements

2.1.1 Cow Detection and Identification

1. The system shall detect cows in real time using fixed camera footage.
2. The system shall identify and separate individual cows using visual tags or appearance-based identification.
3. The system shall count the total number of cows in the camera's field of view.

2.1.2 Tracking and Behavior Monitoring

4. The system shall track each cow's movement across frames.
5. The system shall detect if an identified cow disappears from the footage.
6. The system shall log movement levels (low, moderate, high) for each cow throughout the day.
7. The system shall monitor behavioral and physical indicators associated with approaching calving.

2.1.3 Estrus and Calving Detection

8. The system shall analyze cow behavior patterns and movement features to detect estrus indicators.
9. When estrus likelihood exceeds a threshold, the system shall notify the user.
10. When calving likelihood exceeds the threshold, the system shall notify the user.

2.1.4 Alerting and Notifications

11. The system shall generate daily activity summaries for farm management use.
12. The system shall push real-time notifications to the mobile app for heat detection, calving prediction, and missing cows.
13. The system shall notify the user via the mobile app when a cow is missing.
14. The system shall allow users to adjust alert thresholds and notification preferences.
15. The system shall notify the user if a camera stream becomes unavailable or disconnected.

2.1.5 Mobile Application Features

16. The app shall present individual cow profiles including recent activity, alerts, and timestamps.
17. The system shall authenticate users before allowing access to the mobile app or farm data.
18. The system shall store user settings such as notification preferences, thresholds, and cow profile information.

2.1.6 Data Storage and Historical Logging

19. The system shall store detection, identification, tracking, and alert data with timestamps.
20. The system shall provide historical logs for estrus events, calving predictions, missing-cow events, and activity levels.

2.1.7 Multi-Camera and Farm Infrastructure Support

21. The system shall accept simultaneous footage from multiple cameras across a barn or field.
22. The system shall synchronize detections across cameras to maintain consistent cow identities.
23. The system shall support the integration of new cameras into existing farms.
24. The system shall support the addition and removal of cows from existing farms.
25. The system shall handle duplicate or conflicting cow detections across multiple cameras.

2.2 Non-Functional Requirements

2.2.1 Usability

1. The mobile app interface must be simple enough for farmers with minimal technical background.
2. Alerts must be clear, color-coded, and include cow ID, timestamp, and reason.
3. The system must allow configuration of notifications and thresholds through intuitive sliders, toggles, and drop-down menus.
4. The mobile dashboard must present historical data using clean charts and tables.

2.2.2 Reliability

1. The system must operate continuously 24/7 without requiring manual restarts under normal conditions.
2. Video processing must tolerate temporary network drops and camera signal loss without losing long-term tracking.
3. Camera stream failures must trigger automated reconnection attempts and error logging.
4. The system must correctly resume identification when previously obscured cows re-enter the camera frame.
5. Notifications must not produce false positives exceeding accuracy limits.
6. Stored data must remain consistent even under unexpected shutdowns or power outages.

2.2.3 Performance

1. Inference should operate at near real-time speed.
2. Notification delivery delay must not exceed 5 seconds after an event is detected.

2.2.4 Supportability

1. The system must allow remote software updates for both the backend and the mobile application.
2. Logs must be accessible for debugging with timestamps, errors, and model outputs.
3. The system must allow adding new detection models or retraining existing ones without altering the core architecture.
4. Configuration files (thresholds, camera parameters, alert timing) must be editable without redeploying the system.
5. Documentation for developers and farm technicians must be provided, including setup steps, troubleshooting, and integration guidelines.

2.2.5 Scalability

1. The architecture must support adding new cameras without significantly affecting throughput.
2. The system must scale to farms with at least 100 cows and at least 10 simultaneous camera streams.
3. The database must handle growth in historical records without increasing query time beyond acceptable limits.

2.2.6 Portability

1. The mobile app must support both iOS and Android.
2. Camera feeds must support multiple formats.

3 Feasibility Discussions

3.1 Market & Competitive Analysis

3.1.1 Comprehensive Market Analysis

While researching the current state of Precision Livestock Farming (PLF), we approached the market analysis from two distinct perspectives. First, we investigated the broad spectrum of challenges facing modern dairy farming to understand the macro-level problems. Second, after discussing with stakeholders and clearly defining the scope of our project, we narrowed down our focus to the specific technological solutions available for reproductive monitoring. Therefore, the first part of this analysis covers the general industry challenges, while the second part focuses on specific competitive tools.

Although there is no single definition for operational inefficiency in farming, it can be simply described as operations that rely heavily on manual labor for tasks that require high biological precision. According to industry research, an efficient farm is defined as one that maximizes the yield per head while minimizing the need for human intervention [1]. Therefore, the challenges of traditional farming systems can be categorized into three distinct pillars: Reproductive Inefficiency, Labor Shortages, and Hardware Constraints. Current solutions in the industry generally attempt to solve one of these issues but often fail to address them as a complete system.

3.1.1.1 Reproductive Inefficiency

Reproductive inefficiency refers to the financial and operational loss resulting from a cow failing to become pregnant at the ideal time. Just as complex code is difficult to maintain, a herd with irregular cycles is difficult to manage financially.

- **The Cost of Error:** A single missed estrus cycle results in a 21-day delay in calving. Research indicates that this error costs a farmer between $200 and $300 per cow due to extended non-productive periods and lost milk production [2].
- **Detection Difficulty:** The primary factor contributing to this inefficiency is the biological nature of estrus. The visible signs, such as standing heat, are brief and often last only 8 to 12 hours. Furthermore, these signs frequently occur at night when no farmers are present. Traditional methods relying on visual observation have a detection rate of less than 50%, which is not viable for modern farm profitability.

3.1.1.2 Labor Shortages & Operational Constraints

Another major problem that arises with maintaining a dairy operation is the persistent lack of skilled labor. Much like the institutional knowledge attrition found in legacy software when developers leave, farms suffer when experienced herdsmen retire or quit.

- **The Experience Gap:** Experienced farmers can identify a sick or fertile cow almost instinctively based on years of practice. However, as the workforce shrinks, new staff lack this hands-on experience.
- **Requirement for Automation:** To compensate for this gap, farms are turning to automation. However, most existing tools are focused on production technology, such as robotic milkers, rather than monitoring systems. There is a significant gap in the market for tools that can watch the herd 24/7 and document behavioral changes automatically, providing a digital safety net for the farmer.

3.1.1.3 Hardware Invasiveness & Maintenance Risks

One common fear among farmers adopting new technology is the hardware liability, which refers to the risk that the technology itself becomes a maintenance burden. This is similar to regression risks in software engineering; fixing one problem like detection should not create a new maintenance problem.

- **Invasive Solutions:** Traditional solutions often rely on swallowed sensors, technically known as boluses. These are invasive to the animal and stressful to administer.

- **Ongoing Maintenance Costs:** Wearable sensors such as collars and ear tags require battery replacements, strap adjustments, and physical handling of the animal. If a sensor fails or falls off, the data stream is broken. This maintenance debt discourages many Small-to-Medium Enterprise (SME) farms from modernizing their operations.

3.1.2 Specific Market Analysis

From the list of technologies available on the market and the requirements of our customers, we have concluded that the core features of our system must be: Non-Invasive Monitoring, Automated Estrus Detection, Calving Prediction, and Visual ID Tracking.

We have narrowed down our competitors to these 5 specific tools that provide these functionalities:

Tool	**Non-Invasive**	**Estrus Detection**	**Calving Alert**	**Individual ID**	**Hardware Type**
Moo-tion	**Yes**	**Yes**	**Yes**	**Yes**	**Standard Cameras**
smaXtec		Yes	Yes		Internal Bolus
CowManager		Yes	Yes	Yes	Ear Sensor
Allflex (SenseHub)		Yes	Yes	Yes	Smart Collar
Cainthus	Yes	Yes		Yes	Industrial Cam
OneCup AI	Yes	Yes	Yes	Yes	Industrial Cam

Table 1: Competitive Feature Matrix

- **smaXtec:** This is the market leader in internal monitoring. It uses a bolus sensor that is swallowed by the cow and resides permanently in the stomach. While it provides medical-grade temperature data ($\pm 0.01°C$), it is highly invasive, and the hardware cannot be recovered [4].
- **CowManager:** This tool uses an ear sensor to track ear temperature and activity. It is particularly effective for monitoring eating and rumination behavior, but the ear tags are prone to being ripped out during head-locking or fighting, leading to lost hardware [5].
- **Allflex (SenseHub):** The industry standard for collar-based monitoring. It uses accelerometers to detect motion associated with heat. While highly accurate, the collars require manual adjustment as young cows grow, creating a significant labor burden for the farmer [6].
- **Cainthus (Ever.Ag):** An enterprise-grade computer vision system. It excels at monitoring feed availability and food bunk management in large industrial barns. However, it requires a complex installation of proprietary cameras and servers, making it too expensive for average-sized farms [7].
- **OneCup AI (BETSY):** The closest direct competitor to our project. They use facial recognition technology to track animals. However, their primary market focus has been on beef cattle operations and general health verification, rather than the specific reproductive cycle management required by dairy farmers [8].

3.2 Academic Analysis

After selecting our project topic, our team inspected many academic papers regarding Precision Livestock Farming (PLF). During our research, the scope and the features of our project evolved multiple times. Conducting this detailed academic research allowed us to see the changing trends in academia about moving away from physical sensors toward non-invasive image processing. Initially, we conducted

research to validate whether Computer Vision in Agriculture was a viable thesis topic. Following that, we shifted our focus to the specific Deep Learning architectures capable of handling complex barn environments.

3.2.1 Importance of Our Topic

From the very start of our project, we realized just how widespread and significant the reproductive efficiency problem was. We realized that a missed estrus cycle is the single largest economic drain on a dairy farm. We found that the U.S. dairy industry loses millions annually due to days open, which refers to the days cows are not pregnant [2]. After ensuring the importance of our topic, we moved on to researching what sort of features we should add to our project so that it could facilitate the farmer's daily workflow. It became apparent that the most lacking part of current solutions was integration; farmers have cameras for security and sensors for health, but no tool that combines them. Even though researchers agreed that non-invasive monitoring was the future, the tools that could accurately identify cows and their behavior simultaneously were surprisingly rare [9].

3.2.2 Proposed Solutions Throughout the Years

After our initial research and our decision to focus on Computer Vision, we analyzed various papers to get a better understanding of how to detect estrus visually.

- **The Shift from Sensors:** Early research focused heavily on accelerometers and pedometers. However, we discovered that these devices often generated false positives due to general herd excitement rather than specific estrus behavior.
- **Object Detection Evolution:** We then examined the progression of object detection models. Early attempts used background subtraction, which failed in dynamic barn lighting. The academic consensus shifted toward CNNs (Convolutional Neural Networks). Specifically, YOLO (You Only Look Once) architectures became the standard for real-time cow detection due to their speed and accuracy in crowded environments [9].
- **Behavioral Analysis:** Identifying the action of mounting is harder than identifying the cow itself. We found several technical papers that detailed that 3D-CNNs and LSTM (Long Short-Term Memory) networks are the state-of-the-art solution. These models analyze a sequence of frames rather than static images, allowing the system to distinguish between a cow simply standing near another versus the specific mounting motion indicative of estrus [10]. This academic insight was a turning point for our project, leading us to adopt a temporal model rather than a static image classifier.

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